Factor-base hyperparameter sweep: yield vs `d_ff` vs `D*` vs cost

Date. 2026-10-14. **Status.** Measured on synthetic instances. Every cell passed every control, and the protocol was frozen before the data (`PROTOCOL.md`, commit `31199a70a`). **No security claim**: this is a parameter-space diagnostic on toy fields. It is not an ECDLP speed claim in ecbench's counted unit, and it says nothing about deployed curves.

Code. `examples/fb_hyperparam_sweep.rs`, run source `31199a70a`. It builds on `descent_lowgamma::FactorSubspace`, `pc_degree_harness::refutation_scan` and `prime_fast::{ic_solve, rho_parallel}`, and modifies none of them. **Data.** `data/sweep.json` holds every measured value, plus the `analysis` and `derived_cost_model` sections that the same binary writes. `data/console.log` is the run log. **Run report.** `reports/run-20261014T1804Z/report.pdf`. **Results PDF.** `REPORT.pdf`.

Question

Index calculus with summation polynomials has one main design knob: the factor base. Making it bigger raises relation yield, but it also makes each decomposition system harder. People usually report one point on that trade-off, sometimes as a first-fall degree. This study puts yield, the first-fall degree `d_ff`, the refutation degree `D*` and the measured per-attempt cost of each decomposition oracle on one grid:

- **Binary arm.** `F_{2^n}` with `n = 8…16`, two-point decompositions (S_3). The factor base is the set of `x`-coordinates in an `F_2`-subspace `V`, with `dim V = n' = 2…6`. There are three basis families: `coordinate` (Gaudry/Diem `span{1,…,z^{n'−1}}`), `random` (a full-rank basis; the null for structure) and `subfield` (`V = F_{2^{n'}}`, `n' | n`; this is the factor base that Frobenius preserves). There are two curve classes: `generic_b` (4 random `b` per cell; the null for curve class) and `koblitz_b1` (`b = 1`; S_3 does not depend on `a`). That makes 208 cells.
- **Prime arm (negative control).** `a = −3` prime-order curves of 20, 24 and 28 bits, with the small-`x` factor base, `m ∈ {2, 3}`, and factor-base sizes 64…16,384. Each configuration completes a whole DLP solve for 3 secrets and is compared with single-thread negation rho on the same curve and secret.

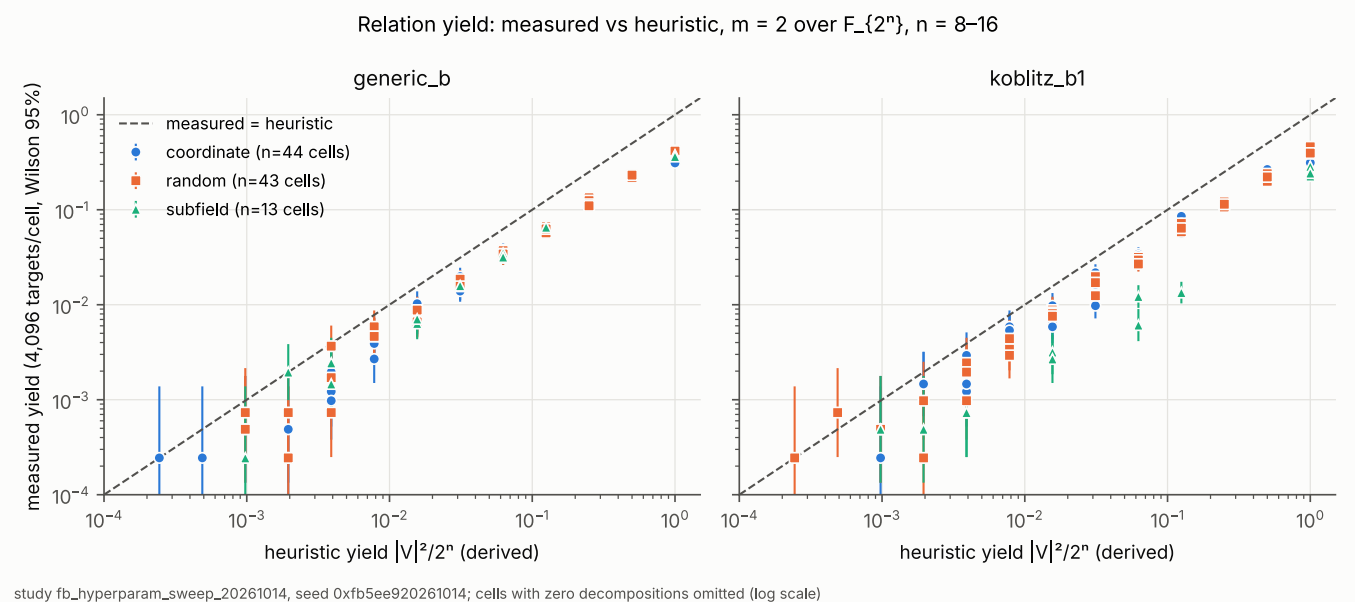
Controls (all 208 cells passed)

control	requirement	result
planted decompositions (positive control)	root finding finds all of them, and brute force finds the first 16	13,312/13,312, and brute force 3,328/3,328
agreement between oracles	root finding = brute force over `V × V`	53,248/53,248 targets agree
calibration of the refutation detector	decomposable targets never refute	0 refuted, out of 4 per cell
censoring	`D*` reached by `d_max = 6`	3,264/3,264 non-decomposable targets refuted, none censored

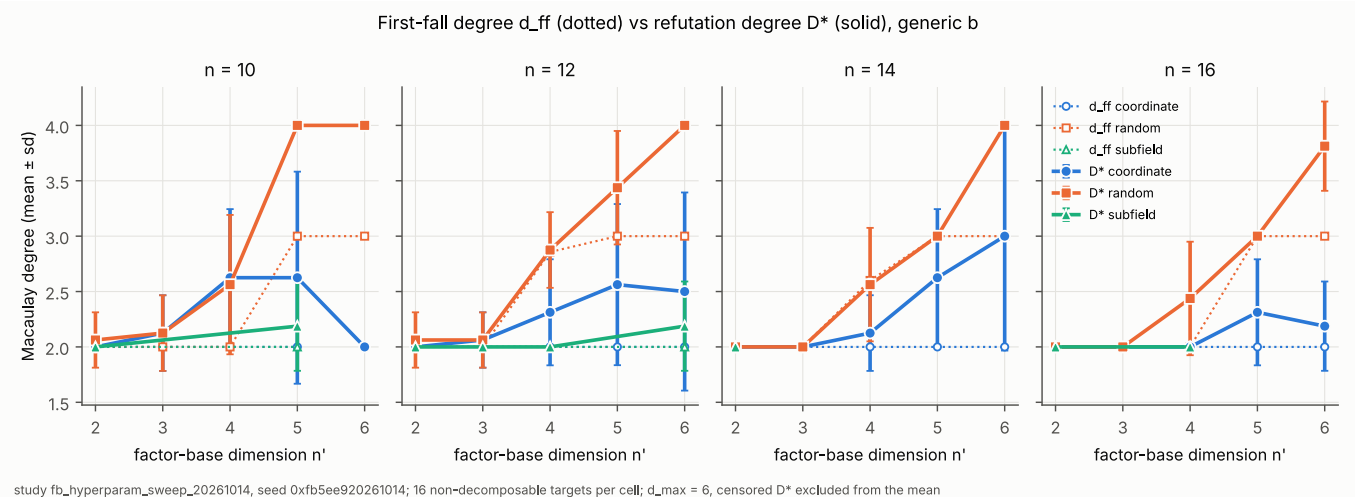
Results

The class of each statement is given in brackets (measured, derived or heuristic), as the `controlled-experiment` skill requires.

H1: the yield heuristic overestimates by about 2×. [measured] The heuristic $|V|^2/2^n$ lies inside the Bonferroni-adjusted Wilson interval in 63 of 208 cells. Excluding subfield cells, the median measured/heuristic ratio is 0.49, 0.51, 0.49, 0.46 and 0.36 at `n' = 2…6`. In the 74 cells with heuristic ≤ 0.05 the median ratio is 0.50. [heuristic explanation, not tested here] For fixed x_1, x_2, S_3 is quadratic in x_3 and has roots in `F_{2^n}` for only about half of the pairs (a trace condition). That gives $\approx |V|^2/2^{n+1}$. Toward `n' = 6` the ratio falls further because the yield saturates (several pairs hit the same target).



H4: `D*` moves with the knobs much more than `d_ff` does. [measured] Across `n'` at fixed `(class, family, n)`, the mean `d_ff` varies by at most 1. For the coordinate and subfield bases it is often exactly 0. Per target, `d_ff` is 2 (1,907 targets) or 3 (950), and is undefined below `D*` for 407 of the 3,264. Over the same ranges the mean `D*` varies by up to 3.0 (coordinate) and 1.6–3.0 (random). This is the point of the sweep: a single first-fall number does not track how hard the system is to solve.

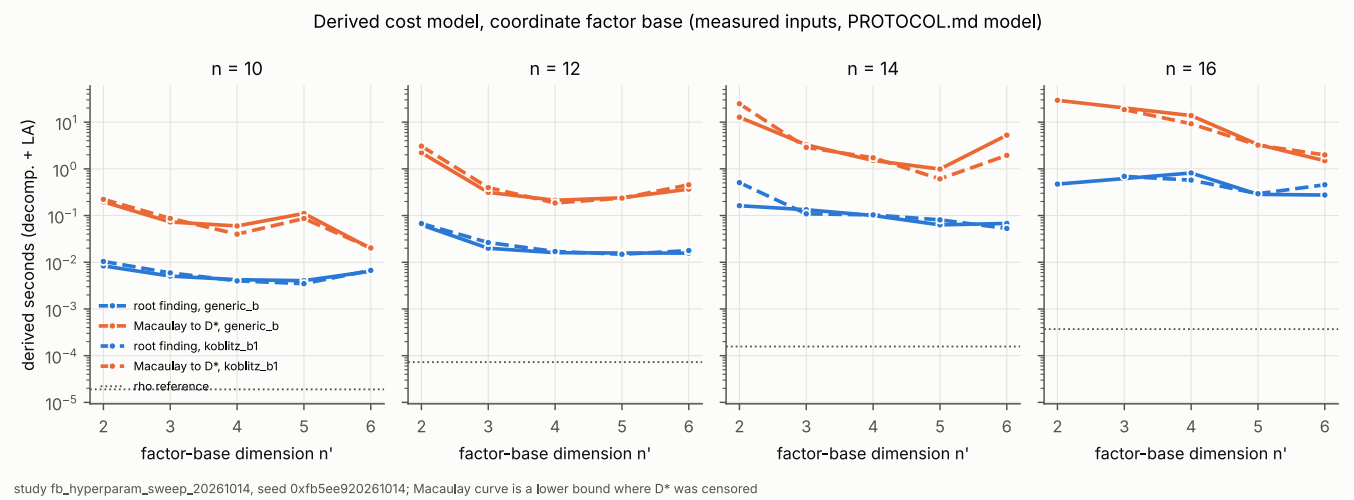


H2: the structure of the basis changes `D*`. [measured, matched pairs, bootstrap intervals, Bonferroni over 118 pairs] The `random` basis has a **higher** `D*` than `coordinate` at matched `(class, n, n')` in 31 of 90 pairs, and lower in none. The differences are +0.4 to +2.0 degrees and concentrate at `n' ≥ 4`. `subfield` vs `coordinate` differs in 1 of 28 pairs (`generic_b`, `n = 15`, `n' = 5`: −0.50, interval [−0.94, −0.13]). Across `n'`, the mean `D*` of the subfield family varies by at most 0.375. This agrees with EXP-E's low-γ/high-γ contrast (`descent_lowgamma`): structured bases refute at low degree and random bases do not.

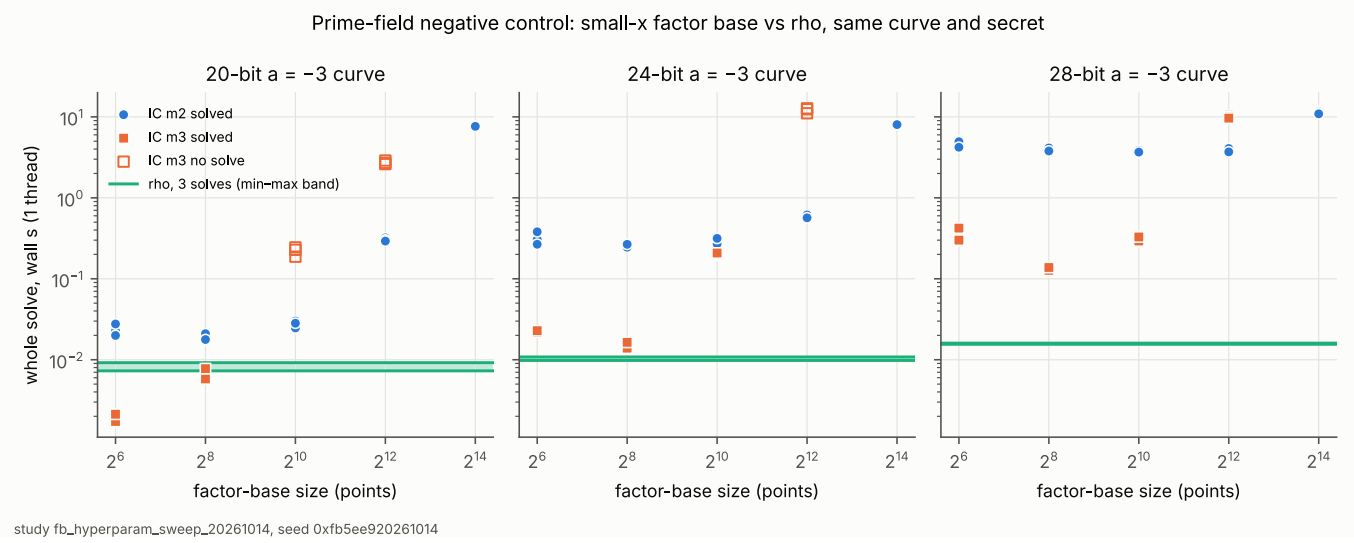
H3: Koblitz `b = 1` vs generic `b`. [measured, Bonferroni over 104 pairs] `D*` differs in **none** of the 104 pairs. Yield does not differ for coordinate bases (0 of 45 lower, 2 higher) or random bases (0 of 45 lower, 3 higher). For the **subfield factor base on a Koblitz curve, yield is lower in 7 of 14 pairs and collapses when `2n' |/ n`**. Examples: `n = 15`, `n' = 5`: 0/4,096 vs 65/4,096; `n = 9`, `n' = 3`: 0.013 vs 0.065; `n = 12`, `n' = 4`: 0.0061 vs 0.0317. [derived, exact argument] When $x_1, x_2 \in F_{2^{n'}}$ and $b \in F_2$, the coefficients of S_3 as a quadratic in x_3 lie in `F_{2^{n'}}`. So every decomposable target lies in $F_{2^{2n'}} \cap F_{2^n} = F_{2^{\gcd(2n',n)}}$, and the yield is at most $2^{\gcd(2n',n)}/2^n$. The cap stays below 1 unless `n = 2n'`. Three of the 14 measured Koblitz × subfield rates sit above the cap. At `n = 14`, `n' = 2` and `n = 15`, `n' = 3` the excess is 1–2 hits out of 4,096 targets, which is within sampling. `n = 12`, `n' = 4` has 25 hits where the cap gives an expected 15 (Poisson tail about 1%). That is the largest deviation, and this study does not follow it up. The Frobenius saving that the cost model credits (relations needed divided by `n'`) does not make up for it. For example, the derived total at `n = 12`, `n' = 4` is 0.051 s for Koblitz vs 0.019 s for generic

b. Only at `n = 2n'` is the cap trivial. Even there the Koblitz yield is 23–40% lower (`n = 8, 10, 12` with `n' = n/2`: 0.286/0.371, 0.226/0.378, 0.243/0.363), although the derived totals come out slightly better (for example 0.0057 s vs 0.0137 s at `n = 12, n' = 6`).

Derived cost model. [derived; inputs measured on this host; model in PROTOCOL.md] Root finding beats Macaulay-to-`D*` per attempt by 3× to 3,900× in each of the 204 cells where Macaulay was timed. Four cells had no non-decomposable target. With root finding, the derived total for the coordinate basis varies across `n'` by a factor of 1.6–16 (median 3.6) per `(class, n)`. The yield rises like $|V|^2$, while the attempt cost and the relations needed each rise like $|V|$. So for `m = 2` the optimum is shallow, with no single sharp interior minimum. The Macaulay curve is not monotone, because `D*` jumps. Every IC configuration with root finding sits 46–6,800× above the rho reference $\sqrt{(\pi 2^n/4)} \cdot$ `t_add`. That is the known result that two-point index calculus on binary curves costs about 2^n targets, which is worse than rho.



H5: the prime negative control. [measured wall time, 1 thread, L0, descriptive] On the 24-bit and 28-bit curves, every index-calculus configuration that solved is slower than the slowest rho solve (fastest IC: 13.7 ms with `m = 3`, FB = 256, at 24 bits, against rho at 10.8 ms; 122.6 ms against 16.0 ms at 28 bits). For `m = 2` it is slower at every size, including 20 bits. **The control's expectation does not hold at 20 bits for `m = 3`**: FB = 64 (1.7 ms) and FB = 256 (5.8 ms) are faster than the slowest rho (9.2 ms). At FB = 1,024 and 4,096 on 20 bits, and FB = 4,096 on 24 bits, `ic_solve` returned no solve in 3/3 runs within 0.2–12.8 s. `ic_solve` returns `None` both when the relation budget runs out and when the linear algebra fails, so the cause is **not diagnosed**. For `m = 3`, the `yield` field counts relations per target and goes above 1. ecbench's counted-unit studies remain the reference for prime index calculus vs rho; this arm only checks that the sweep's instrument sees the expected ordering at non-toy sizes.



Limits

- Only `m = 2`. The interior optimum in the original question needs `m ≥ 3` (S_4, `binary_semaev_s4`, `examples/ffd_s4_subspace.rs`). That is the next arm.

- `n ≤ 16` and `n' ≤ 6` are toy sizes, chosen so that every `D*` refutes by `d_max`. `d_ff` uses the operational definition in `pc_degree_harness`.
- "Decomposable" means S_3 has a zero with $x_1, x_2 \in V$. It does not check that x_1, x_2 lift to points over F_{2^n} (the same convention as the existing harness). The yields are therefore upper bounds on relations from rational points.
- Wall times come from one shared 4-vCPU Xeon at 2.10 GHz, L0 isolation, a single run per cell. The cost model is derived and not admitted as a speed measurement.
- The Frobenius credit in the cost model (divide by `n'`) is a modelling assumption, and nothing here verifies it.

Reproduce

```
cargo run --release --example fb_hyperparam_sweep -- research/fb_hyperparam_sweep_20261014/data
python3 research/fb_hyperparam_sweep_20261014/plot.py   # matplotlib; plots recorded values only
```

The run took 291 s of wall time on the host above (`data/console.log`).